# Custom AMI Creation — Task Steps & Observations :

## Task:

1. **Launch Ubuntu EC2 Instance**
   - Type: t2.micro (Free Tier)
   - OS: Ubuntu 22.04 LTS
2. **Install Apache Web Server**

```
sudo apt update
sudo apt install apache2 -y
sudo systemctl start apache2
sudo systemctl enable apache2
```

3. **Test Web Server**
   - Open browser: `http://<Public-IP>` → Should display: *Welcome to Custom AMI Page*
4. **Create Custom AMI**
   - Actions → Image → Create Image → Name: `Ubuntu-Apache-AMI`
5. **Launch New EC2 From AMI**
   - New instance launched from custom AMI should already have Apache & index.html
   - Confirm via browser again.

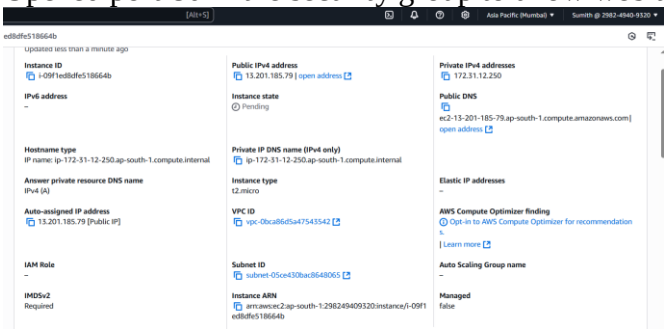
## What is an AMI?

An **Amazon Machine Image (AMI)** is a **template** that contains the software configuration (OS + app + environment + EBS snapshots) required to launch an EC2 instance. Think of it as a reusable blueprint.

## Steps Performed in Detail:

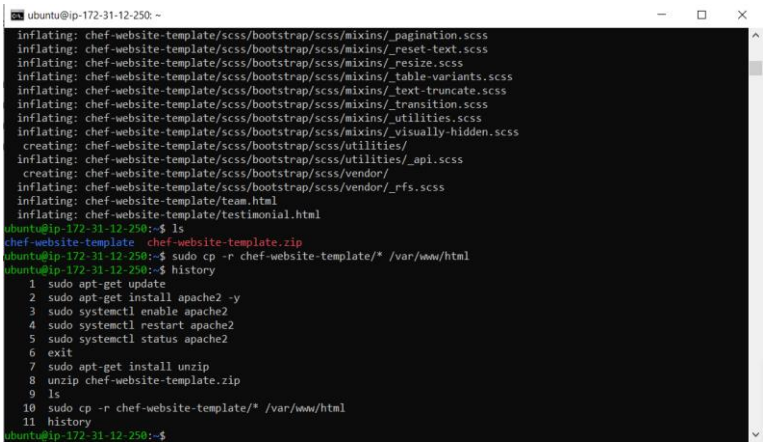
### *Step 1: Launch Ubuntu EC2*

- Used Ubuntu 22.04 LTS, t2.micro in ap-south-1a (Mumbai).
- Opened port **80** in the security group to allow web traffic.

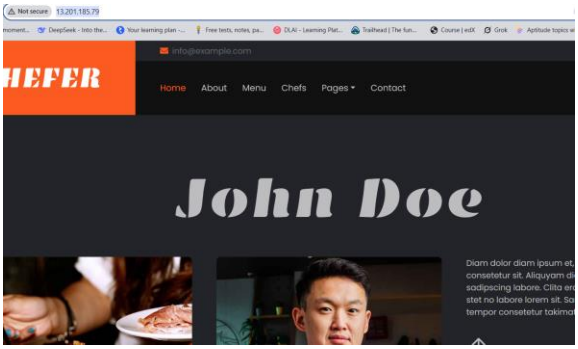


### *Step 2: Install Apache and Host a Page*

```
sudo apt update
sudo apt install apache2 -y
sudo systemctl start apache2
sudo systemctl enable apache2
```

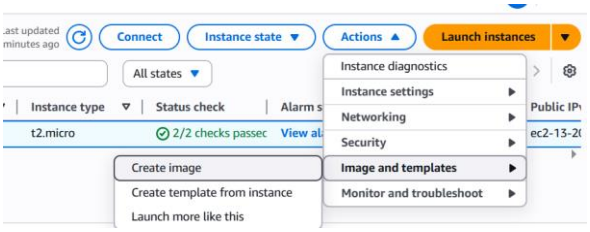


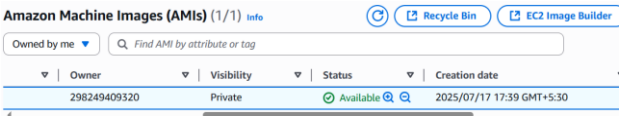

- This sets up a basic HTTP web server.
- Confirm via browser using public IP.




### *Step 3: Create Custom AMI*

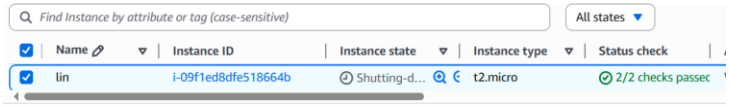
- Go to EC2 → Instances → Select Instance → Actions → Create Image.
- AMI includes **root volume snapshot** + metadata (architecture, instance type, kernel).
- Wait until image status = `available`.

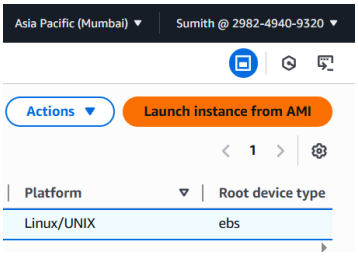


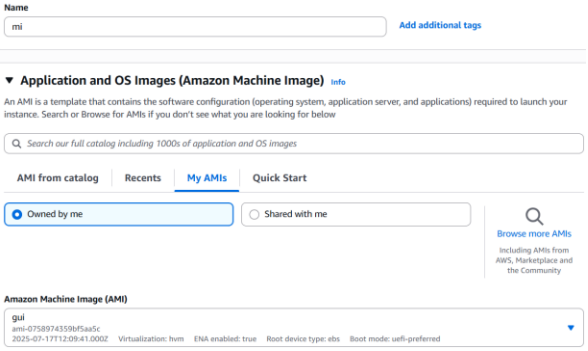



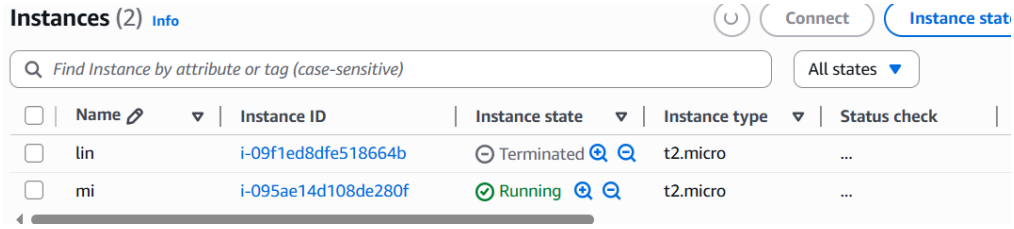

## *Step 4: Launch New EC2 from AMI*

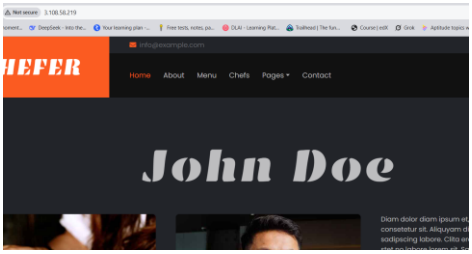
- When launched, the instance booted up with Apache already running and custom index.html already in place.



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### ⚙ What Happens Internally?

- AWS creates an EBS **snapshot** of the root volume.
- Stores metadata (e.g., launch permissions, block device mapping).
- When reused, AMI can launch **identical EC2 setups**.

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### ▯ Best Practices

- Use **custom AMIs** for scaling apps with pre-installed software.
- Combine with **Auto Scaling Groups** for HA architectures.
- Periodically update base AMIs for patching and compliance.

# Snapshot Creation and Recovery Using Windows EC2 Instance (GUI & CLI)

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## ✅ PART 1: Create Snapshot from Windows EC2

### Step 1: Launch Windows EC2 Instance

- Use: Windows Server 2019 or 2022 Base AMI
- Instance type: `t2.micro`
- Enable RDP in security group (Port 3389)

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### Step 2: Add & Attach New EBS Volume

1. Go to **EC2 → Volumes → Create Volume**
   - Size: 1–2 GiB
   - AZ: Same as your Windows EC2 (e.g., `ap-south-1a`)
2. Attach Volume:
   - Actions → Attach → Select your Windows EC2
   - Device Name: Use suggested default (e.g., `/dev/sdf`)

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### Step 3: Login to Windows EC2

- Use RDP with public IP + administrator password
- Launch: **Disk Management**
  - (Right-click Windows logo → Disk Management)

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### Step 4: Initialize & Format Volume

1. Right-click the new disk → **Initialize Disk** (GPT)
2. Right-click unallocated space → **New Simple Volume**
3. Format it as **NTFS** and assign a drive letter (e.g., `E:`)

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### Step 5: Create Dummy File

- Open Notepad → Type: `This is a snapshot test`
- Save as `test.txt` inside `E:\` or your new drive

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### Step 6: Create Snapshot

1. Go to AWS Console → EC2 → **Volumes**
2. Select the volume (attached one)
3. Actions → **Create Snapshot**
    - Name: `WindowsSnapshotTest`
    - Description: Snapshot for recovery test

⏳ Wait for snapshot status = `completed`.

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## ✅ PART 2: Restore Volume from Snapshot

### Step 7: Delete Volume (Simulate Disaster)

- Detach and delete original volume (Volume with `test.txt`)

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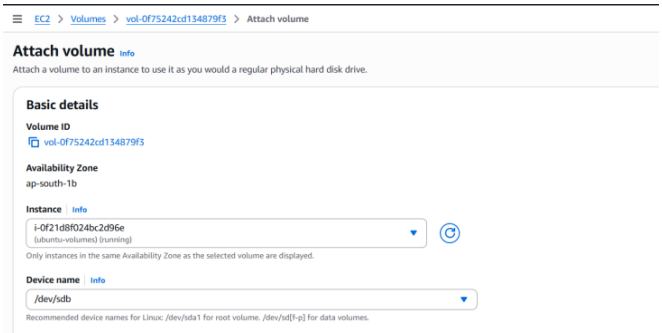
### Step 8: Create New Volume from Snapshot

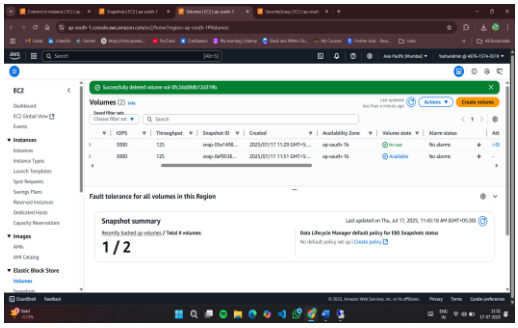
1. Go to EC2 → Snapshots → Select snapshot
2. Actions → **Create Volume**
    - AZ: Same as EC2 instance
    - Size: Same or larger than original
3. Attach it to the EC2 instance (e.g., as `/dev/sdf` again)

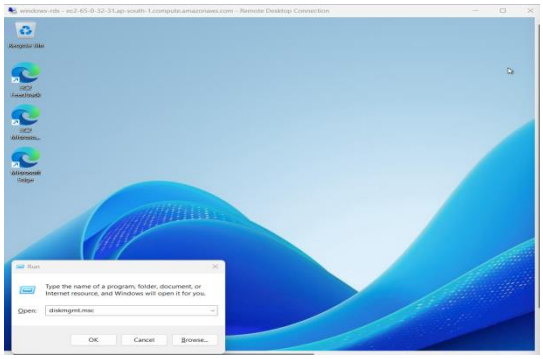
---

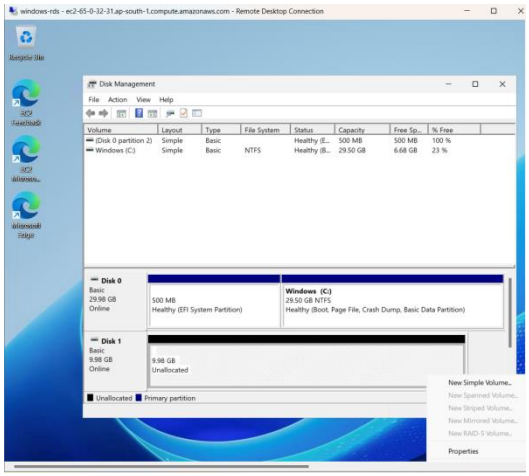
### Step 9: Verify Recovery in Windows

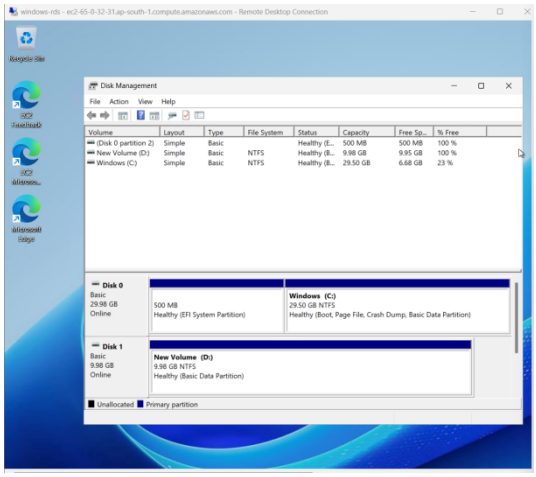
1. RDP into EC2 again
2. Open **Disk Management**
3. If the drive does not show automatically:
    - Right-click → **Rescan Disks**
    - Assign drive letter if missing
4. Open `E:\test.txt` (or your assigned letter)

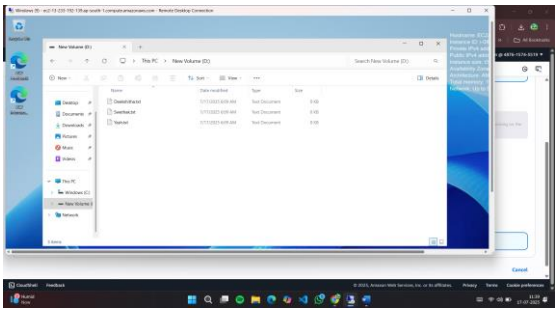

EC2 > Volumes > vol-0f75242cd134879f3 > Attach volume
Attach volume Info
Attach a volume to an instance to use it as you would a regular physical hard disk drive.
Basic details
Volume ID
vol-0f75242cd134879f3
Availability Zone
ap-south-1b
Instance Info
i-0f21d8f024bc2d96e
(ubuntu-volumes) (running)
Only instances in the same Availability Zone as the selected volume are displayed.
Device name Info
/dev/sdb
Recommended device names for Linux: /dev/sda1 for root volume. /dev/sd[f-p] for data volumes.

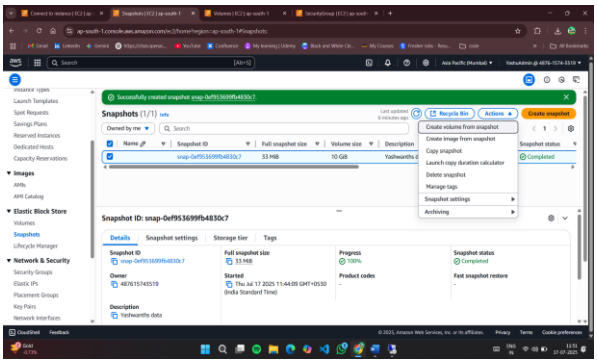

Volumes (2)
Fault tolerance for all volumes in this Region
Snapshot summary
1 / 2

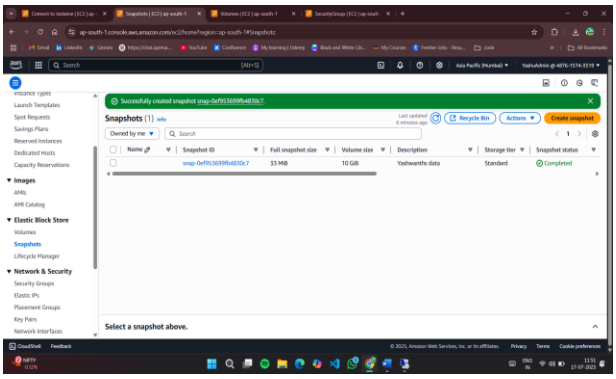

Run
Type the name of a program, folder, document, or Internet resource, and Windows will open it for you.
Open: diskmgmt.msc
OK
Cancel
Browse...

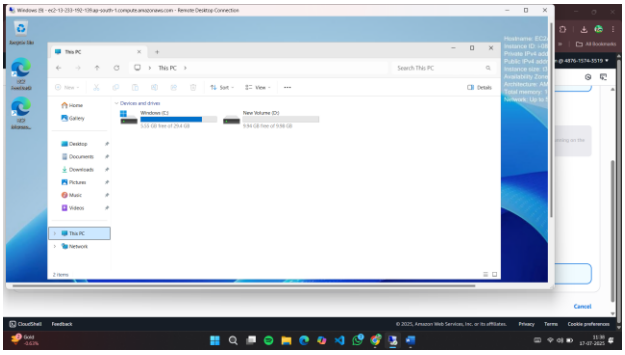

windows-rds - ec2-65-0-32-31.ap-south-1.compute.amazonaws.com - Remote Desktop Connection
Disk Management
File Action View Help
Volume Layout Type File System Status Capacity Free Sp... % Free
(Disk 0 partition 2) Simple Basic Healthy (E... 500 MB 500 MB 100 %
Windows (C:) Simple Basic NTFS Healthy (B... 29.50 GB 6.68 GB 23 %
Disk 0
Basic
29.98 GB
Online
500 MB
Healthy (EFI System Partition)
Windows (C:)
29.50 GB NTFS
Healthy (Boot, Page File, Crash Dump, Basic Data Partition)
Disk 1
Basic
9.98 GB
Online
9.98 GB
Unallocated
Unallocated Primary partition
New Simple Volume...
New Spanned Volume...
New Striped Volume...
New Mirrored Volume...
New RAID-5 Volume...
Properties


windows-rds - ec2-65-0-32-31.ap-south-1.compute.amazonaws.com - Remote Desktop Connection
Disk Management
File Action View Help
Volume Layout Type File System Status Capacity Free Sp... % Free
(Disk 0 partition 2) Simple Basic Healthy (E... 500 MB 500 MB 100 %
New Volume (D:) Simple Basic NTFS Healthy (B... 9.98 GB 9.95 GB 100 %
Windows (C:) Simple Basic NTFS Healthy (B... 29.50 GB 6.68 GB 23 %
Disk 0
Basic
29.98 GB
Online
500 MB
Healthy (EFI System Partition)
Windows (C:)
29.50 GB NTFS
Healthy (Boot, Page File, Crash Dump, Basic Data Partition)
Disk 1
Basic
9.98 GB
Online
New Volume (D:)
9.98 GB NTFS
Healthy (Basic Data Partition)
Unallocated Primary partition





## Billing Internals:

- **Snapshots** billed per GB/month.
- **Unattached EBS volumes** billed per hour.
- **Elastic IPs** billed when idle.
- AWS does **not automatically delete** unused resources.

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## ▯ Best Practices:

- Regularly schedule **monthly cleanup review**.
- Use **resource tagging** (`env=dev`, `owner=you`) for cost tracing.
- Set **budgets and alerts** via AWS Budgets.
- Use **AWS Trusted Advisor** for cost-saving recommendations.

# AMI vs Snapshot — Deep Technical Comparison:

| Feature | AMI (Amazon Machine Image) | Snapshot (EBS) |
|---|---|---|
| Scope | Full EC2 instance setup (OS + app + volume mappings) | Backup of a single EBS volume |
| Purpose | Quickly replicate EC2 instance environments | Data backup, versioning, and disaster recovery |
| Includes OS? | ✅ Yes | ❌ No (only data from EBS) |
| Launchable? | ✅ Yes, can launch EC2 instances directly | ❌ No. Must create volume → attach to EC2 |
| Speed of Use | Immediate deployment of full servers | Manual recovery, slower setup |
| Storage Type | Backed by EBS snapshots, plus metadata | Stored in S3 (internally), block-level incremental backups |
| Use Case | Template for autoscaling, environment replication, golden images | Data protection, rollback, disaster recovery |
| Automation | Can be baked with tools like **Packer, EC2 Image Builder** | Can be scheduled with **Data Lifecycle Manager** |

## AWS Account Suspension — Policy & Recovery

### When Does AWS Suspend Your Account?

If you **don't pay bills**, AWS moves your account from `ACTIVE` → `SUSPENDED`.

### ⏱ Grace Period:

- Up to **30 days** after bill due.
- After 30–45 days, account may be **restricted**:
  - No EC2 launch
  - API access blocked
  - Running resources might stop

### ❌ What Gets Affected?

- EC2s stop
- Data might still be charged (S3, EBS, snapshots)
- New deployments blocked

## 🔄 Recovery Steps:

1. **Login and Pay Outstanding Balance**
    - Use credit/debit card or UPI
2. **Contact AWS Support (Billing)**
    - If auto reactivation fails
    - Provide payment confirmation
3. **Reactivate Manually**
    - AWS may take 24–48 hours to fully reactivate services

## ⚠️ Risk:

After 90+ days of suspension, account and resources can be **permanently deleted**.